

Apex Diagnostics

Comprehensive Pre-Marital Health Report

Clinical Profile: Male IDA

Name: Mr. Karan Malhotra	Age/Sex: 27 Yrs / M	Date: 02 Jun 2026, 03:50 PM	ID: ADX-19267
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BLOOD GROUP - ABO TYPING			
Parameter	Result	Reference	Unit
Blood Group (A / B / AB / O)	O	-	-

BLOOD GROUP - RH (RHESUS) TYPING			
Parameter	Result	Reference	Unit
Rh Factor (Positive / Negative)	Positive	-	-

COMPLETE BLOOD COUNT (CBC) / HEMOGRAM			
Parameter	Result	Reference	Unit
Total Red Blood Cell Count (RBC)	4.83	4.5-5.9	mill/μL
Hemoglobin (Hb)	8.6	13.5-17.5	g/dL
Hematocrit / Packed Cell Volume (PCV)	33.0	41.0-53.0	%
Mean Corpuscular Volume (MCV)	68.3	80.0-100.0	fL
Mean Corpuscular Hemoglobin (MCH)	31.1	27.0-33.0	pg
Mean Corpuscular Hemoglobin Concentration (MCHC)	26.1	32.0-36.0	g/dL
Red Cell Distribution Width - CV (RDW-CV)	14.1	11.5-14.5	%
Red Cell Distribution Width - SD (RDW-SD)	41.1	39.0-46.0	fL
Total White Blood Cell Count (WBC / TLC)	7.4	4.5-11.0	thou/μL
Neutrophils (%)	60.5	40-80	%
Neutrophils - Absolute Count	4.48	2.0-7.0	thou/μL
Lymphocytes (%)	33.8	20-40	%
Lymphocytes - Absolute Count	2.22	1.0-3.0	thou/μL
Monocytes (%)	6.0	2-10	%
Monocytes - Absolute Count	0.44	0.2-1.0	thou/μL
Eosinophils (%)	3.0	1-6	%
Eosinophils - Absolute Count	0.22	0.02-0.5	thou/μL
Basophils (%)	0.5	0-2	%
Basophils - Absolute Count	0.04	0.0-0.1	thou/μL
Immature Granulocytes (%)	0.2	0-0.5	%
Immature Granulocytes - Absolute Count	0.01	0-0.03	thou/μL
Nucleated Red Blood Cells (NRBC) - Absolute Count	0.0	0.0	thou/μL
Nucleated Red Blood Cells (NRBC) %	0.0	0.0	%
Platelet Count	204	150-450	thou/μL
Mean Platelet Volume (MPV)	9.5	7.5-11.5	fL
Platelet Distribution Width (PDW)	12.0	9.0-17.0	fL
Plateletcrit (PCT)	0.25	0.15-0.35	%
Platelet-Large Cell Ratio (P-LCR)	25.0	15.0-35.0	%

ERYTHROCYTE SEDIMENTATION RATE (ESR)			
Parameter	Result	Reference	Unit
Erythrocyte Sedimentation Rate (ESR)	10	< 15	mm/hr

BLOOD SUGAR - FASTING (FBS / FBG)			
Parameter	Result	Reference	Unit
Fasting Blood Glucose (FBG)	82.2	70-100	mg/dL

BLOOD SUGAR - RANDOM (RBS / BSR)			
Parameter	Result	Reference	Unit
Random Blood Sugar (RBS)	104.1	70-140	mg/dL

GLYCATED HEMOGLOBIN (HBA1C)			
Parameter	Result	Reference	Unit
HbA1c (%)	5.2	4.0-5.6	%
Estimated Average Glucose (eAG)	102.5	70-126	mg/dL

AVERAGE BLOOD GLUCOSE (ABG)			
Parameter	Result	Reference	Unit
Average Blood Glucose (ABG)	97.2	< 117	mg/dL

LIVER FUNCTION TEST (LFT) - TOTAL BILIRUBIN			
Parameter	Result	Reference	Unit
Total Bilirubin	0.84	0.3-1.2	mg/dL
Direct (Conjugated) Bilirubin	0.21	< 0.3	mg/dL
Indirect (Unconjugated) Bilirubin	0.21	< 0.3	mg/dL

LIVER FUNCTION TEST (LFT) - SGPT / ALT			
Parameter	Result	Reference	Unit
Alanine Aminotransferase (SGPT / ALT)	20	< 41	U/L

LIVER FUNCTION TEST (LFT) - SGOT / AST			
Parameter	Result	Reference	Unit
Aspartate Aminotransferase (SGOT / AST)	21	< 40	U/L

KIDNEY FUNCTION TEST (KFT)			
Parameter	Result	Reference	Unit
Serum Creatinine	0.83	0.6-1.2	mg/dL
Blood Urea Nitrogen (BUN)	11	7-20	mg/dL
BUN / Creatinine Ratio	13.3	10-20	Ratio
Serum Uric Acid	6.1	3.5-7.2	mg/dL
Serum Urea	24	15-40	mg/dL
Estimated Glomerular Filtration Rate (eGFR)	107	> 90	mL/min
Serum Sodium (Na ⁺)	140	135-145	mEq/L
Serum Potassium (K ⁺)	4.2	3.5-5.1	mEq/L
Serum Chloride (Cl ⁻)	102	98-107	mEq/L
Serum Bicarbonate (HCO ₃ ⁻)	26	22-29	mEq/L

SERUM IRON			
Parameter	Result	Reference	Unit
Serum Iron	95	50-170	μg/dL

TOTAL IRON BINDING CAPACITY (TIBC)			
Parameter	Result	Reference	Unit
Total Iron Binding Capacity (TIBC)	494	250-450	μg/dL

UNSATURATED IRON BINDING CAPACITY (UIBC)			
Parameter	Result	Reference	Unit
Unsaturated Iron Binding Capacity (UIBC)	399	110-370	μg/dL

TRANSFERRIN SATURATION (%)			
Parameter	Result	Reference	Unit
Transferrin Saturation (%)	19	20-50	%

SERUM FERRITIN			
Parameter	Result	Reference	Unit
Serum Ferritin	5	20-250	ng/mL

HEMOGLOBIN HPLC / VARIANT ANALYSIS			
Parameter	Result	Reference	Unit
Hemoglobin A (HbA)	97.1	> 95.0	%
Hemoglobin A2 (HbA2)	2.4	1.5-3.5	%
Hemoglobin F (HbF / Fetal Hemoglobin)	0.5	< 2.0	%
Hemoglobin S (HbS)	0.0	0.0	%
Hemoglobin C (HbC)	0.0	0.0	%
Hemoglobin D (HbD)	0.0	0.0	%
Hemoglobin E (HbE)	0.0	0.0	%
Total Hemoglobin (g/dL)	8.6	12.0-16.0	g/dL
Unknown / Other Hemoglobin Fractions	0.0	0.0	%
Interpretation / Diagnosis	Normal Study	Normal Study	-

HEMOGLOBIN A2			
Parameter	Result	Reference	Unit
Hemoglobin A2 (HbA2) %	2.4	1.5-3.5	%

HEMOGLOBIN C			
Parameter	Result	Reference	Unit
Hemoglobin C (HbC) %	0.0	0.0	%

HEMOGLOBIN D			
Parameter	Result	Reference	Unit
Hemoglobin D (HbD) %	0.0	0.0	%

HEMOGLOBIN F			
Parameter	Result	Reference	Unit
Hemoglobin F / Fetal Hemoglobin (HbF) %	0.5	< 2.0	%

HEMOGLOBIN S			
Parameter	Result	Reference	Unit
Hemoglobin S (HbS) %	0.0	0.0	%

THYROID STIMULATING HORMONE (TSH)			
Parameter	Result	Reference	Unit
Thyroid Stimulating Hormone (TSH / uTSH)	2.02	0.4-4.0	µIU/mL
Triiodothyronine (T3) - Total	120	80-200	ng/dL
Thyroxine (T4) - Total	8.0	5.0-12.0	µg/dL
Free T3 (FT3)	3.2	2.3-4.2	pg/mL
Free T4 (FT4)	1.1	0.9-1.7	ng/dL

VITAMIN B12 (CYANOCOBALAMIN)			
Parameter	Result	Reference	Unit
Vitamin B12 (Cobalamin) - Serum	555	200-900	pg/mL

LIPID PROFILE / LIPID SCREEN			
Parameter	Result	Reference	Unit
Total Cholesterol	146	< 200	mg/dL
High-Density Lipoprotein Cholesterol (HDL-C)	49	> 40	mg/dL
Low-Density Lipoprotein Cholesterol (LDL-C)	94	< 100	mg/dL
Very Low-Density Lipoprotein (VLDL-C)	24	2-30	mg/dL
Triglycerides	77	< 150	mg/dL
Non-HDL Cholesterol	97	< 130	mg/dL
Total Cholesterol / HDL Ratio (TC:HDL)	3.0	< 5.0	Ratio
LDL / HDL Ratio	1.9	< 3.0	Ratio

SYPHILIS - VDRL / RPR			
Parameter	Result	Reference	Unit
VDRL / RPR Result (Reactive / Non-Reactive)	Non-Reactive	Non-Reactive	-
RPR Titre (if reactive)	Not Applicable	Not Applicable	-

HEPATITIS B SURFACE ANTIGEN (HBSAG)			
Parameter	Result	Reference	Unit
HBsAg - Qualitative Result (Reactive / Non-Reactive)	Non-Reactive	Non-Reactive	-
HBsAg - OD Value / Signal-to-Cutoff Ratio	0.2	< 0.9	S/CO

HEPATITIS C VIRUS ANTIBODY (ANTI-HCV)			
Parameter	Result	Reference	Unit
Anti-HCV Antibody - Qualitative Result (Reactive / Non-Reactive)	Non-Reactive	Non-Reactive	-
Anti-HCV - OD Value / Signal-to-Cutoff Ratio	0.1	< 0.9	S/CO

HIV 1 & 2 ANTIBODY TEST			
Parameter	Result	Reference	Unit
HIV 1 & 2 Antibody - Result (Reactive / Non-Reactive)	Non-Reactive	Non-Reactive	-
HIV ICT (Immunochromatographic Test) Result	Non-Reactive	Non-Reactive	-

HIV COMBO (P24 ANTIGEN + ANTIBODY) TEST			
Parameter	Result	Reference	Unit
HIV P24 Antigen - Result (Detected / Not Detected)	Not Detected	Not Detected	-
HIV 1 & 2 Antibody - Result (Reactive / Non-Reactive)	Non-Reactive	Non-Reactive	-
HIV Combo Assay - OD Value / S/CO Ratio	0.1	< 0.9	S/CO
HIV 1 Differentiation Result	Negative	Negative	-
HIV 2 Differentiation Result	Negative	Negative	-

URINE ROUTINE & MICROSCOPY EXAMINATION			
Parameter	Result	Reference	Unit
Colour	Pale Yellow	Pale Yellow	-
Appearance / Clarity / Turbidity	Clear	Clear	-
Specific Gravity	1.015	1.005-1.030	-
pH	6.0	4.5-8.0	-
Urinary Protein	Absent	Absent	-
Urinary Glucose	Absent	Absent	-
Urinary Ketones	Absent	Absent	-
Urinary Bilirubin	Absent	Absent	-
Urobilinogen	Normal	Normal	-
Blood / Haemoglobin in Urine	Absent	Absent	-
Nitrites	Absent	Absent	-
Leukocyte Esterase	Absent	Absent	-
Pus Cells (WBCs) per HPF	1-2	0-5	per HPF
Red Blood Cells (RBCs) per HPF	0-1	0-2	per HPF
Epithelial Cells	Rare	Rare	per HPF
Casts	21	< 40	U/L
Crystals	Absent	Absent	-
Bacteria	Absent	Absent	-
Yeast / Fungi	21	< 40	U/L
Mucus Threads	Absent	Absent	-
Amorphous Material	6.0	4.5-8.0	-

SEMEN ANALYSIS

Parameter	Result	Reference	Unit
Semen Volume	4.1	>= 1.5	mL
Semen pH	7.5	>= 7.2	-
Colour / Appearance	Pale Yellow	Pale Yellow	-
Viscosity	Normal	Normal	-
Liquefaction Time	25	< 60	mins
Sperm Concentration	57	>= 15	mill/mL
Total Sperm Count per Ejaculate	234	>= 39	mill/ ejac.
Total Sperm Motility (%)	59	>= 42	%
Progressive Motility (PR) %	56	>= 30	%
Non-Progressive Motility (NP) %	56	>= 30	%
Immotility (IM) %	41	-	%
Total Motile Sperm Count (TMSC)	138	-	million
Sperm Morphology - Normal Forms (%)	6.0	4.5-8.0	-
Sperm Head Defects (%)	60	-	%
Sperm Midpiece Defects (%)	20	-	%
Sperm Tail Defects (%)	11	-	%
Sperm Vitality / Viability (%)	75	>= 54	%
Pus Cells / Leucocytes (per HPF)	1-2	0-5	per HPF
Round Cells per mL	< 1	< 1	mill/mL
Sperm Agglutination	Absent	Absent	-
Interpretation / Fertility Prognosis	Normozoospermia	Normozoospermia	-